Report (5510 words)

Chapter 1: Introduction (804 words)

When computing researchers conduct a literature review, they often need to read many papers and identify the research methodology of each one. This project treats methodology as four extractable roles: technical method, task, dataset, and evaluation metric. Identifying these four components for each paper is useful for comparing related work and for understanding how methods in a field have changed over time. When reviewing many papers, however, this process is slow and manual.

Consider "Attention Is All You Need" [D6], a well-known machine learning paper. The title signals attention mechanisms but does not describe how the research was conducted. That information is spread across the paper and requires careful reading to extract. The paper's methodology, once identified, can be summarized as follows:

Methodology role	Component
Technical method	Transformer
Research task	machine translation
Evaluation dataset	WMT (Workshop on Machine Translation) datasets
Evaluation metric	BLEU (Bilingual Evaluation Understudy) score

Table 1: Role-based methodology profile for "Attention Is All You Need" [D6].

A reader needing this summary would currently have to read through the paper and construct it themselves.

1. Topic labels and role profiles

Topic classifiers and search tools help researchers find papers about a subject. But a topic label answers "what is this paper about?", not "how was this research conducted?" Two papers on the same topic can use different methods, train on different datasets, and report different metrics. A role-based profile answers the second question. It does not replace reading; it supports the first pass through a set of papers — before a researcher decides which ones to read in depth.

The distinction matters in practice. A student reviewing papers on text classification needs to know not only which papers address that topic, but also which ones use transformer models versus older approaches, which datasets are most commonly used, and which metrics are reported. Without a structured profile, this comparison requires reading each paper.

2. Project goal and approach

This project addresses **Template 12.1** from the Natural Language Processing (NLP) module: Identifying research methodologies used in computing research. The project builds a prototype pipeline that takes a computing paper PDF and returns sentences grouped by four methodology roles. The profile contains four fields:

- **TechnicalMethod** — the method, algorithm, or architecture the authors used (e.g. Transformer)
- **Task** — the problem the research addresses (e.g. machine translation)
- **Dataset** — the data used for training or evaluation (e.g. WMT datasets)
- **EvaluationMetric** — the metric used to assess results (e.g. BLEU score)

The pipeline works in three stages. First, a PDF is converted to TEI (Text Encoding Initiative) XML by GROBID [7], which identifies sections with headings. Second, sections are filtered (References and Related Work are excluded), and each remaining sentence is cleaned and validated. Third, a zero-shot Natural Language Inference (NLI) model assigns one of the four roles to each sentence without requiring task-specific training data.

I use zero-shot classification because I do not have a labelled corpus for this task. Supervised approaches require substantial annotation effort: Jain et al. [5] built a comparable information extraction system using 438 annotated papers and four expert PhD-level annotators. I target a broader range of computing papers (including systems, algorithms, human-computer interaction (HCI), and ML research) where no comparable annotated dataset is available. Yin et al. [11] show that NLI-based zero-shot classification can assign arbitrary labels to text without task-specific training examples, making it suitable for this setting.

Research design (e.g. experiment vs survey) [6] and philosophical worldview are out of scope. These components can be subjective and may not appear as explicit phrases in the paper text. The four roles listed above appear more directly in the text and are more tractable to classify automatically at the sentence level.

3. Target users

The primary users are computing students who need to survey multiple papers for a literature review or research project. They need to find the method, task, dataset, and evaluation metric quickly, before reading a paper in full. The system is intended to support this first pass, not to replace reading.

The output is designed to be inspectable. Each role field contains actual sentences from the paper, allowing the user to judge whether a classified sentence is genuinely relevant to the paper's methodology. At this prototype stage, the output is full sentences rather than short extracted terms. This limitation is discussed in Chapter 4.

4. Report structure

Chapter 2 reviews three areas of related work: how research methodology in computing is defined and structured; how methodology-related information can be extracted from scientific papers; and how zero-shot classification can assign labels to text without labeled training data. The chapter identifies a gap this project investigates.

Chapter 3 describes the system design. It covers the four-role schema and its justification, the full pipeline architecture, the choice of technologies, the evaluation approach, and the work plan.

Chapter 4 presents the feature prototype. It describes the implementation of the zero-shot NLI classification step, demonstrates results on six computing papers, evaluates the prototype against manually identified gold labels, and discusses planned improvements for the next iteration.

Chapter 2: Literature Review (1841 words)

Chapter 1 showed a four-role profile for "Attention Is All You Need" [D6]. Figure 1 shows a fuller view of the same paper, including the design strategy and data generation method defined by Oates [9].

```
Methodology:
    Design or strategy: design and creation + experiment
    Data generation method: documents
    Technical method: Transformer
    Task: machine translation
    Dataset: WMT machine translation datasets
    EvaluationMetric: BLEU score
```

Figure 1: Research Methodology from "Attention Is All You Need" [D6].

This review focuses on three areas: how methodology is defined, how information is extracted from papers, and how classification can work without training data. At the end, this review will identify a gap: existing methods can extract some information from papers, but extracting these methodology components consistently from computing papers is still difficult.

1. Defining Research Methodology

Research methodology in computing papers can be described using a structured vocabulary, but defining it is not the same as extracting it.

Oates [9] provides six research strategies (experiment, design and creation, survey, case study, action research, and ethnography) and four data generation methods (interviews, observations, questionnaires, and documents). His book defines the vocabulary that researchers use to describe their methodology in papers, so my project needs these concept names to identify what to extract. His book was published in 2006, but it still provides useful categories for describing how computing researchers conduct their work. However, the six strategies were designed for human researchers to self-classify their own work — papers rarely contain the explicit phrase "this is an experiment". The vocabulary can be used to name what to look for, but it may not transfer directly to automatic extraction from text.

Pilkington & Pretorius [10] go further: they formalize the structure using UML (Unified Modeling Language) and ontology engineering, with the goal of "providing clear and unambiguous semantics" [10] — a formal structure, not a textbook description. Key concepts are ResearchScheme, PhilosophicalWorldview, ResearchDesign, and ResearchMethod. ResearchScheme belongs to one PhilosophicalWorldview, has one or more ResearchDesigns, and has one or more ResearchMethods. The paper tries to solve the problem that students and supervisors had no shared, consistent vocabulary for methodology, so they often used the same terms with different meanings.

A philosophical worldview is one of the key components in Pilkington & Pretorius [10], but it tends not to appear as an explicit phrase in completed research papers — this appears to be an empirical pattern rather than just a design decision. I therefore exclude it from extraction.

Oates [9] gives concept names. Pilkington & Pretorius [10] give formal relationships between those concepts. My project uses vocabulary from Oates and formal structure from Pilkington & Pretorius.

Both works are designed for human use. Neither provides a system to extract methodology components automatically from text. These works suggest that methodology can be defined and formalized. The question is whether any system can extract it.

2. Closest Prior Work

Systems that extract methodology-like entities from papers exist [2, 3], but the closest supervised approaches require labeled training data that I do not have.

Jain et al. [5] extract four entity types: Dataset, Metric, Task, and Method.

```
Dataset: WMT machine translation datasets
Metric: BLEU score
Task: machine translation
Method: Transformer
```

Figure 2: Entity types from "Attention Is All You Need" [D6].

These four types closely match the four roles in this project. This supports using these four roles in the prototype.

Jain et al. [5] operate at the document level. The authors argue that "a significant amount of information can only be gleaned from analyzing the full document" [5] — relations may span sections, not just sentences.

But Jain et al. [5] required 438 annotated papers and 4 expert PhD-level annotators (Cohen-κ 95%). The corpus comes from Papers with Code, which covers only ML benchmarks. My project targets general computing papers (systems, algorithms, HCI, etc.) and has no annotated corpus. Two constraints make the Jain et al. [5] approach difficult to adopt directly: the corpus is limited to ML benchmark papers, and annotation required 4 PhD-level experts at Cohen-κ 95% — neither condition is available for this project.

Ma et al. [8] propose a metric-driven mechanism schema that extracts three components (mechanism, task, and metric) from NLP papers using a query-guided sequence-to-sequence model, but their work is limited to the NLP domain and does not extend to general computing research.

Ghosh et al. [2, 3] use supervised transformer-based sequence labeling to extract methodology component names from AI research papers. They argue that methodology names are difficult to extract because they are large, fast-evolving, domain-specific, and context-dependent. Unlike Jain et al. [5], who extract four entity types at the document level, Ghosh et al. focus narrowly on TechnicalMethod. This makes direct reuse difficult: even if the sequence labeling approach transferred, it would not address Task, Dataset, or EvaluationMetric — three of the four roles in this project. The approach is also specific to AI research papers; the training data does not cover general computing research such as systems, algorithms, or HCI. Like Jain et al. [5], the approach is supervised and requires annotated training data that this project does not have.

Färber et al. [1] extract methods and datasets from scientific publications using domain-specific named entity recognition (NER) followed by usage classification, which distinguishes whether each mention is used by the authors or only cited as prior work. This "used vs mentioned" distinction is directly relevant to a noise problem in the prototype: a sentence in the BERT Introduction describing ELMo's approach entails "technical method" under NLI but belongs to a different paper. However, Färber et al. [1] cover only Method and Dataset, not Task or

EvaluationMetric. Their NER model also requires labeled entity mentions, which are not available for this project.

Across these three supervised systems, a common pattern emerges. Jain et al. [5] achieve the broadest coverage (all four roles at document level) but require 438 annotated papers from a single domain. Ghosh et al. [2, 3] narrow the scope to TechnicalMethod and apply finer-grained sequence labeling, but cover only one of the four roles. Färber et al. [1] add a useful usage signal for two roles, but do not extend to Task or EvaluationMetric. All three depend on domain-specific labeled corpora. For a project targeting general computing papers where no annotated methodology corpus appears to exist, reusing any of these directly would require rebuilding the annotation infrastructure from scratch. This constraint motivates a zero-shot approach, which removes the labeled data requirement at the cost of domain adaptation.

These papers support the entity types, but their supervised methods depend on annotation that this project does not have. A zero-shot method is therefore a reasonable direction.

3. Zero-shot Classification

Zero-shot NLI can assign roles to text without task-specific training data, but applying it to scientific papers introduces a domain mismatch risk.

Yin et al. [11] define zero-shot text classification as assigning a label to text without any task-specific training examples.

Yin et al. [11] show that NLI can classify text into many possible labels by turning the label into a natural language hypothesis (for example, "this text is about [label]") and asking a model whether the text entails it. No labeled examples for the target labels are needed.

aspect	**labels**	**interpretation**	**example hypothesis (word)**	**example hypothesis (wordnet definition)**
topic	sports etc.	this text is about ?	"?"= sports	"?" = an active diversion requiring physical exertion and competition
emotion	anger etc.	this text expresses ?	"?"= anger	"?" = a strong emotion; a feeling that is oriented toward some real or supposed grievance
situation	shelter etc.	The people there need ?	"?"= shelter	"?" = a structure that provides privacy and protection from danger

Table 2 (reproduced from Yin et al. [11]): example hypotheses for three task types.

This is relevant to my classification step: assigning one of four methodology roles to each sentence without a labelled corpus. I apply the same entailment approach with four methodology roles:

role	hypothesis (this project, short label)
TechnicalMethod	technical_method
Task	task
Dataset	dataset
EvaluationMetric	evaluation_metric

Table 3: hypothesis set used in this project (short label format, selected by hypothesis set comparison).

For TechnicalMethod, a longer hypothesis was also tested: "This text describes a technique, algorithm, system, or architecture used or proposed in the research." The hypothesis set comparison investigates whether this verbose form can extract TechnicalMethod more accurately than the short label on scientific text.

However, a domain mismatch risk exists. Yin et al. test on Yahoo News articles, emotion tweets, and crisis situation reports. Their NLI model is trained on MNLI (Multi-Genre Natural Language Inference; covers news, fiction, and telephone speech). None of these are scientific papers, which tend to use dense technical vocabulary, passive constructions, and section-based structure. The model may not handle scientific text well — this risk is built into the model weights, not just a test-time shift.

I test this risk by comparing short and verbose hypothesis labels on scientific text.

The supervised approaches reviewed in Section 2 address domain mismatch more directly, because each was trained on annotated scientific text from the target domain. Yin et al. [11] avoid this annotation requirement by relying on the semantic relationship between text and label hypothesis. The trade-off is concrete: the supervised systems achieve strong extraction within their training domains but do not transfer to a new domain without re-annotation. Zero-shot NLI shifts the bottleneck from annotation to hypothesis design. The hypothesis comparison in this project tests this directly: verbose_v1 assigned 94% of BERT sentences (244 of 258) to EvaluationMetric — a single poorly chosen hypothesis collapsed the distribution. Short labels gave the best probe score (3/4) and the most balanced distribution across four roles. This shows that hypothesis quality is a real design cost in zero-shot systems, not merely a minor parameter choice.

Zero-shot NLI reduces the labeled data requirement. The question is whether any prior work combines this approach with a methodology schema on scientific papers.

4. Synthesis

It is difficult to find existing work that applies zero-shot NLI with a structured methodology schema to general computing papers. This motivates the prototype.

Section 2 established the basis for a structured approach. Oates [9] and Pilkington & Pretorius [10] suggest that research methodology has formal, structured components, but neither work names the specific extraction roles used in this project. The four roles (TechnicalMethod, Task, Dataset, and EvaluationMetric) draw more directly on Jain et al. [5]. Both Oates and Pilkington are designed for human use. Neither provides a system to extract methodology from text automatically.

Section 3 showed that extraction of the same four types is possible. Jain et al. [5] built a working system, but it required 438 annotated papers, 4 PhD-level annotators, and a corpus limited to ML benchmarks. This approach does not readily generalize to general computing papers without similar annotation effort.

Section 4 showed that zero-shot NLI removes the labeled data requirement. Yin et al. [11] demonstrate that NLI can classify text into any label without task-specific training. But their approach was tested only on news articles, tweets, and crisis reports — not scientific papers. Domain mismatch remains an open risk.

I could not find prior work that combines these elements in the same way: the structured methodology concept from Oates and Pilkington, the four-role vocabulary from Jain et al., the zero-shot NLI method from Yin et al., and application to general computing papers. I apply Yin et al.'s entailment approach with the 4-role schema on GROBID-parsed computing papers, without requiring annotated data. This made the approach worth testing in a prototype.

The three supervised systems in Section 2 confirm that the four-role extraction task is achievable with sufficient annotation. Jain et al. [5] extract all four roles but require ML-benchmark annotation at scale. Ghosh et al. [2, 3] narrow to TechnicalMethod. Färber et al. [1] add usage sensitivity for two roles. None of the conditions that make these approaches work (large labeled corpora, single-domain scope) apply to this project. Yin et al.'s [11] zero-shot approach removes the annotation requirement but was tested only on general-domain text. My project occupies a specific position in this space: it draws on the schema from Jain et al. [5], the extraction method from Yin et al. [11], the preprocessing from GROBID [7], and the domain vocabulary from Oates [9] and Pilkington & Pretorius [10]. The prototype tests whether this combination is effective in practice.

Chapter 3: Design (1380 words)

The system extracts research methodology from computing papers. An input is a PDF, and an output is a role-based profile (see Figure 1 in Chapter 2 for an example).

1. Domain and Users

The domain is computing research papers. The main targets are systems, ML, algorithm, and HCI.

The primary users are computing students doing literature reviews (see Chapter 1, "Target users"). Secondary users may include early-stage researchers or supervisors who want a quick overview of a paper. The output is designed mainly for students: simple, inspectable, and based on sentences from the original paper rather than hidden model decisions.

2. Design Justification

Jain et al. [5] needed 438 annotated papers and 4 PhD-level annotators for a supervised approach. I have no annotated corpus. Yin et al. [11] show that NLI can classify text into many possible labels without task-specific training. Zero-shot NLI is suitable for this prototype because it avoids task-specific training data.

The role vocabulary draws on two sources serving different functions. Oates [9] and Pilkington & Pretorius [10] suggest that research methodology has formal, structured components, providing a concept-level justification for extracting them. The specific four roles in this project (TechnicalMethod, Task, Dataset, and EvaluationMetric) draw more directly on Jain et al. [5] (SciREX), who annotated the same four types (Dataset, Metric, Task, Method) across 438 papers, providing evidence that these categories are recognizable in paper text.

Research design (e.g. experiment vs. survey) is subjective — two readers can assign different labels to the same paper. A philosophical worldview may not appear in the paper text at all. The four roles are more likely to appear as surface text than research design or worldview.

The hypothesis is that sentences in a paper tend to describe one role at a time. A sentence about the dataset does not also describe the method. If this holds, sentence-level classification may be sufficient. I tested this on six papers: Transformer [D6], BERT [D3], AlexNet [D5], ResNet [D4], MapReduce [D2], and Google Search [D1]. Results appear to support the assumption for ML papers. Systems papers (MapReduce, Google Search) showed weaker fit because they do not follow the standard ML benchmark structure.

A sentence like "The feature-based approach, such as ELMo, applies independently trained context representations" appears in the Introduction of BERT and describes another paper's method, not BERT's. Skipping the Related Work section by heading removes some noise, but not sentences in the Introduction that describe prior work. A second NLI step with labels ["used by the authors", "mentioned as prior or related work"] may reduce this noise at the sentence level without needing more keyword rules [1]. This may reduce some prior-work sentences in the TechnicalMethod output.

An early run with verbose hypotheses sent almost all BERT sentences to EvaluationMetric, which forced a comparison of four hypothesis sets before settling on short labels.

This design matches the intended use case: the system is not intended to replace expert reading, but to support the first pass of a literature review by showing likely methodology sentences that the user can inspect and verify.

3. Overall Structure

The pipeline runs as follows: PDF → GROBID (runs locally via Docker) → TEI XML → section filtering (skip References, Acknowledgements, Related Work by heading) → sentence splitting with spaCy + pre_clean() + is_valid() → role NLI (TechnicalMethod / Task / Dataset / EvaluationMetric) with threshold 0.5 → MethodologyProfile output as JSON. The next iteration changes the extraction unit from sentence-level classification to document-level extraction (see Chapter 4, Section 5).

```
PDF
  ↓
GROBID (Docker)
  ↓
TEI XML
  ↓
Section filtering
  ↓
Sentence splitting + cleaning
  ↓
Zero-shot NLI role classification
  ↓
MethodologyProfile JSON
```

Figure 3: Overall pipeline of the proto2 system.

The input is a PDF of a computing research paper. After GROBID, the intermediate format is TEI XML. Each section has a heading attribute and body text. The abstract is separate from the body sections.

The output is a MethodologyProfile with four fields: TechnicalMethod, Task, Dataset, EvaluationMetric. Each field is a list of sentences from the paper. Final output is a Python dict printed as JSON. Example:

```json
{
  "TechnicalMethod": ["We propose the Transformer..."],
  "Task": ["applied to machine translation..."],
  "Dataset": ["on the WMT 2014 English-German..."],
  "EvaluationMetric": ["achieving 28.4 BLEU on the WMT 2014..."]
}
```

Figure 4: Example output of the system (proto2 prototype).

The current prototype accepts all sentences above a threshold, which can result in a large number of output sentences per role (e.g. 100+ for TechnicalMethod). This is a known prototype limitation.

GROBID runs locally via Docker because it is a Java server process (~1 GB). The NLI pipeline runs on Google Colab because it needs a GPU for fast inference and installs Python packages (transformers, torch, spacy). The TEI XML produced by GROBID is uploaded to Colab manually.

4. Key Technologies and Methods

GROBID [7] converts a PDF into TEI XML, dividing the paper into sections with headings and body text. Without GROBID, the input would be raw PDF text with no section boundaries, making it difficult to filter by section (e.g. skip References or Related Work).

The prototype uses `cross-encoder/nli-deberta-v3-small` [4] (~300 MB, fits free Colab GPU). Zero-shot is therefore suitable for this prototype because: (1) I have no labelled corpus for this task, so supervised training is not straightforward; (2) Yin et al. show NLI can classify into many possible labels without task-specific training.

Two pre-processing steps clean each sentence before classification. (1) `pre_clean()` strips inline citation markers like [13] or [4, 27] using a regex. Citations break sentence boundaries and cause the splitter to produce short fragments. (2) `is_valid()` drops sentences shorter than 30 characters or without at least one real word. This removes citation stubs (e.g. "[4, 27, 28]"), bullet symbols, and other formatting artefacts that would produce noisy classifications.

Four hypothesis sets (short and three verbose variants) were tested on the BERT paper. Short labels gave the best probe score and the most balanced role distribution. Full results and analysis are in Chapter 4, Section 2.

A later iteration plans to use a long-context LLM to extract one term per role from the cleaned full paper text, with one supporting sentence for each term.

5. Work Plan

The work plan is shown visually in Appendix A as a Gantt chart.

Period	Main task	Output
Before 29 June	Complete all chapters, prototype, and video	Preliminary Report
Iteration 1 (post-submission)	Replace sentence-level classification with document-level extraction using a long-context LLM; return one term and one supporting sentence per role	Short-form methodology profile
Final stage	Compare NLI prototype and document-level extraction results; analyse failures across paper types; write Final Report	Final submission

Table 4: Work plan summary.

The major tasks are:

- Done: background research, literature notes (Oates, Pilkington, Jain, Yin, GROBID, CSO), pitch
- Done: proto2 prototype — GROBID pipeline, NLI classification, section filtering, pre-processing, hypothesis comparison
- Done: Chapter 1 — Introduction
- Done: Chapter 2 — Literature Review
- Done: Chapter 3 — Design
- Done: Chapter 4 — Feature Prototype
- Done: demonstration video (MP4, 3–5 min)
- In progress: Preliminary Report submission (29 June)
- Post-submission: document-level extraction prototype (proto3), extended evaluation
- Final stage: testing, analysis, Final Report writing and submission

The Preliminary Report submission deadline is 29 June. See Appendix A (Figures A1–A4) for the full project roadmap.

6. Test and Evaluation

For each paper × role, the system checks whether any accepted sentence (score ≥ 0.5) contains the gold label as a substring. A role is correct (○) if at least one classified sentence contains the gold label. Incorrect (×) otherwise.

The gold labels for the three test papers are shown in Table 5.

Paper	TechnicalMethod	Task	Dataset	EvaluationMetric
Transformer	Transformer	machine translation	WMT	BLEU
BERT	BERT	GLUE (General Language Understanding Evaluation) / SQuAD (Stanford Question Answering Dataset)	BooksCorpus / Wikipedia	accuracy / F1
AlexNet	AlexNet	image classification	ImageNet	top-1 / top-5 error

Table 5: Gold labels — 3 papers × 4 roles.

Results are presented as a table: rows = 4 roles, columns = 3 papers, each cell = ○ (correct) or × (wrong). Total = 12 data points. The matching sentence and score are shown for each ○ cell to make the result inspectable.

Success is defined as ≥ 10 of 12 correct. 10/12 suggests the system finds relevant sentences for most roles across most papers. Lower than 8/12 would indicate a systematic problem worth investigating.

The evaluation is intentionally small but inspectable. Each paper-role pair is judged by whether the system retrieves at least one relevant sentence containing the expected gold label. To avoid hiding errors behind large outputs, the evaluation will also report the number of accepted sentences per role and show the top matching sentence with its score. This makes it possible to see both whether the system finds the correct evidence and whether the output is too broad.

Known constraints of this approach: only 3 papers (too small for statistical claims); substring match is loose; systems papers (MapReduce [D2], Google Search [D1]) do not fit the 4-role structure and are excluded; gold labels were written by the author with no formal inter-annotator agreement. An extended evaluation covering all 6 papers is in Appendix B.

The analysis will identify which role or paper type failed and explain why (e.g. EvaluationMetric is hardest to capture; systems papers have no standard dataset). Chapter 4 describes the limitations of the current approach and the planned direction for the next iteration.

Chapter 4: Feature Prototype (1485 words)

The prototype takes a TEI XML file produced by GROBID from a computing research paper and classifies each sentence by research methodology role using zero-shot NLI to produce a JSON object with four lists — TechnicalMethod, Task, Dataset, and EvaluationMetric.

Zero-shot NLI role assignment is the central design choice this chapter evaluates (see Chapter 3, Section 2 for justification).

1. Implementation

1.1 Preprocessing

The pipeline applies two preprocessing steps before classification.

First, `pre_clean()` removes inline citation markers such as `[13]` or `[4, 27]` using a regex. Without this step, spaCy may split a sentence at the bracket, producing broken fragments. For example:

> `"The model outperforms [4, 27] the baseline."` →
> `"The model outperforms the baseline."`

Second, `is_valid()` drops sentences that are shorter than 30 characters or contain no word of three or more letters. This removes bullet characters, lone numbers, and citation stubs that pass through sentence splitting but carry no information. After these two steps, the Transformer paper produced 183 valid sentences for classification.

1.2 Section Filtering

References, Acknowledgements, and Related Work are excluded.

- References and Acknowledgements are excluded by exact heading match (`SKIP_HEADINGS`). These sections list citations and credits, not methodology.
- Related Work is excluded by keyword match (`SKIP_KEYWORDS`). Subsections are also skipped automatically by tracking the `n` attribute (e.g. if Related Work is `n="2"` , then `n="2.1"` and `n="2.2"` are also skipped). The reason is that Related Work describes other papers' methods, which the NLI model classifies as TechnicalMethod of the target paper. Testing on BERT showed that excluding Related Work reduced TechnicalMethod from 67 to 62 sentences (−5).

All other body sections are included: Abstract, Introduction, Method/Architecture, Dataset, Experiment, Results, Conclusion, etc. An earlier version filtered to sections whose heading matched keywords such as "experiment" or "result", but the `Training Data` section in the Transformer paper has neither keyword, and the WMT dataset was missed entirely. Switching to all body sections improved Dataset recall significantly.

2. Demonstration

The prototype was tested on six papers. Table 6 shows the number of accepted sentences per role.

Paper	TechnicalMethod	Task	Dataset	EvaluationMetric	Notes
Transformer	62	14	4	3	short: balanced
BERT	62	23	15	13	short: balanced
AlexNet	51	6	11	4	short: balanced
ResNet	51	6	14	12	short: balanced
MapReduce	151	24	3	5	short: TM heavy, weak DS/EM
Google Search	69	21	8	29	short: no standard benchmark

Table 6: Accepted sentences per role for six papers.

All six papers were run with short labels. ML papers (Transformer, BERT, AlexNet, ResNet) produced balanced output across all four roles. Systems papers (MapReduce, Google Search) produced very large TechnicalMethod counts and few Dataset or EvaluationMetric sentences, consistent with their lack of standard ML benchmark structure.

2.1 Hypothesis Set Comparison

The choice of hypothesis text has a large effect on classification. Four hypothesis sets were tested on the BERT paper (258 sentences). A probe set of four known-answer sentences (one per role) was used to measure correctness.

Set	Probe (4 gold labels)	TechnicalMethod	Task	Dataset	EvaluationMetric
short	**3/4**	124	70	33	31
verbose_v1	2/4	11	0	3	244

Set	Probe (4 gold labels)	TechnicalMethod	Task	Dataset	EvaluationMetric
verbose_v2	2/4	63	11	19	165
verbose_v3	2/4	131	56	60	11

Table 7: Hypothesis set comparison on the BERT paper (258 sentences).

Verbose hypotheses introduced strong label bias: verbose_v1 and verbose_v2 sent nearly all sentences to EvaluationMetric, while verbose_v3 shifted the bias to TechnicalMethod. Short labels gave the best probe score (3/4) and the most balanced distribution.

3. Evaluation

3.1 Method

The evaluation follows the approach designed in Chapter 3, Section 6: a gold label check where a role is correct (○) if any accepted sentence contains the gold label as a substring. Jain et al. [5] used a similar role-based recall check in SciREX, evaluating whether predicted spans match the annotated entity per role. This evaluation is recall-oriented: it tests whether the expected gold label is retrieved somewhere in the accepted sentences, but it does not measure how much irrelevant text is also returned. A result of 10/12 means the gold label appears in at least one accepted sentence per role-paper pair, not that the full output is a clean methodology profile.

3.2 Results

Gold labels used (based on the planned labels in Chapter 3, Table 5, refined after running the pipeline):

Paper	TechnicalMethod	Task	Dataset	EvaluationMetric
Transformer	"Transformer"	"machine translation"	"WMT"	"BLEU"
BERT	"BERT"	"GLUE"	"BooksCorpus"	"F1"
AlexNet	"convolutional"	"object recognition"	"ImageNet"	"top-5"

Table 8: Gold labels as used in evaluation (AlexNet TechnicalMethod refined to "convolutional" since the name "AlexNet" was coined after publication).

Result (○ = gold label found in any accepted sentence, ☓ = not found):

Paper	TechnicalMethod	Task	Dataset	EvaluationMetric
Transformer	○	☓	○	☓
BERT	○	○	○	○
AlexNet	○	○	○	○

Table 9: Gold label evaluation results (10/12). Extended results for all 6 papers are in Appendix B.

3.3 Analysis

BERT scored ○ on all four roles, which suggests that the pipeline can find all types of methodology information in a well-structured ML paper using short labels. AlexNet also scored ○ on all four roles when using "convolutional" as the TechnicalMethod gold label (the 2012 paper does not use the name "AlexNet" — that name was coined later).

The Transformer scored ☓ on Task and EvaluationMetric. On Task, 14 sentences were accepted but none contains "machine translation"; the key sentence ("Experiments on two machine translation tasks show the model to be superior") was classified as TechnicalMethod (score 0.53). On EvaluationMetric, 3 sentences were accepted but none contains "BLEU"; metric sentences such as "Our model achieves 28.4 BLEU" were classified as Task. Dataset is ✓ because 4 sentences were accepted and one contains "WMT". TechnicalMethod tends to be the easier role because it appears in dedicated sections with explicit mentions. Task appears to be the most difficult role: it is often stated implicitly or in sentences that the model assigns to another role. An extended evaluation covering ResNet, MapReduce, and Google Search is in Appendix B.

The 10/12 result is better interpreted as an upper bound on recall than as a precision or accuracy claim: it shows the system retrieves at least one relevant sentence for 10 of 12 role-paper pairs, but does not indicate whether the remaining output is clean or useful. The AlexNet gold label was also revised after running the pipeline, changing it from "AlexNet" to "convolutional", to match the terminology the 2012 paper actually uses; this represents evaluator influence on the result and limits how strongly the score can be generalised.

Because the intended use is first-pass literature review support, output volume also matters. A result is less useful if the correct term appears only inside a very large set of accepted sentences. I therefore treat the number of accepted sentences per role (Table 6) as a usability signal alongside the binary correct/incorrect result. By this measure, ML papers perform better than systems papers: BERT and AlexNet retrieve the gold label within 4–15 accepted sentences per role, while MapReduce produces 151 TechnicalMethod sentences — too many to inspect in a first pass.

4. Limitations

Four types of noise were observed.

The first type is Introduction noise. Introduction sections describe other papers' methods, which the NLI model incorrectly classifies as TechnicalMethod of the target paper. For example, "The feature-based approach, such as ELMo..." scored 0.87 as TechnicalMethod in the BERT paper. Excluding the Introduction on BERT reduced TechnicalMethod from 62 to 54 sentences, but also removed correct sentences about BERT itself, so full exclusion risks losing signal.

The second type is quoted or example text. Text that is quoted or used as an example in the paper body is classified as a real claim. For example, "you looked at a lot of pages from my Web site." from the Google Search paper [D1] was classified as Task.

The third type is GROBID artefacts. Author contribution text can appear in the GROBID abstract element, producing irrelevant sentences in the TechnicalMethod output.

The fourth type is large output volume. MapReduce produced 151 TechnicalMethod sentences, because there is no upper limit on accepted sentences.

5. Improvements for the Next Iteration

Sentence-level classification produced useful evidence sentences, but it did not produce a clean methodology profile. MapReduce returned 151 TechnicalMethod sentences, with no principled way to select the primary one. A deeper problem is that classifying sentences in isolation cannot reliably distinguish the paper's own method from methods cited in related work. This is not only a threshold problem: even a better threshold would still treat each sentence independently.

For this reason, the next iteration changes the extraction unit from sentence-level classification to document-level extraction. Jain et al. [5] argue that methodology extraction often requires document-level context because relevant information may be spread across sections. The next prototype will therefore use a long-context LLM to read the cleaned paper text and return one term per role as JSON. Each term must be supported by a quoted sentence from the paper, so the output remains inspectable.

6. Technical Challenge

The prototype raised three technical issues with zero-shot NLI classification on academic text.

The first challenge is that hypothesis engineering is non-trivial. More detailed label descriptions do not necessarily improve accuracy: verbose_v1 and verbose_v2 sent nearly all BERT sentences to EvaluationMetric (244 of 258), while verbose_v3 shifted the bias to TechnicalMethod. Short labels achieved the best probe score and most balanced distribution, but this required four iterations of hypothesis design to discover.

The second challenge is domain mismatch. The NLI model was trained on general-domain benchmarks, but academic writing is structurally different: sentences are longer, more technical, and contain citation markers and figure references that were not in the training data. The model classifies these without any task-specific training on scientific text.

The third challenge is authorship attribution. The sentence "The feature-based approach, such as ELMo..." correctly entails "technical method" according to the NLI model (it does describe a method), but the method belongs to a different paper. Distinguishing a paper's own methods from those it cites was not handled well by the current NLI step, since it requires understanding who the authors are and what the paper claims.

References

[1] Michael Färber, Alexander Albers, and Felix Schüber. 2021. Identifying Used Methods and Datasets in Scientific Publications. In *Proceedings of the Second Workshop on Scholarly Document Understanding (SDU@AAAI 2021)*. https://ceur-ws.org/Vol-2831/paper19.pdf

[2] Madhusudan Ghosh, Debasis Ganguly, Partha Basuchowdhuri, and Sudip Kumar Naskar. 2023a. Enhancing AI research paper analysis: methodology component extraction using factored transformer-based sequence modeling. arXiv:2311.03401. https://doi.org/10.48550/arXiv.2311.03401

[3] Madhusudan Ghosh, Debasis Ganguly, Partha Basuchowdhuri, and Sudip Kumar Naskar. 2023b. Extracting methodology components from AI research papers: a data-driven factored sequence labeling approach. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (CIKM 2023)*. https://doi.org/10.1145/3583780.3615258

[4] Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2021. DeBERTaV3: Improving DeBERTa using ELECTRA-Style Pre-Training with Gradient-Disentangled Embedding Sharing. arXiv:2111.09543. https://doi.org/10.48550/arXiv.2111.09543

[5] Sarthak Jain, Madeleine Van Zuylen, Hannaneh Hajishirzi, and Iz Beltagy. 2020. SciREX: A Challenge Dataset for Document-Level Information Extraction. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Online, July 2020. Association for Computational Linguistics, 7506–7516. https://doi.org/10.18653/v1/2020.acl-main.670

[6] Zsolt T. Kosztyán, Tünde Király, Tibor Csizmadia, Attila Imre Katona, and Ágnes Vathy-Fogarassy. 2025. Automated research methodology classification using machine learning. *Engineering Applications of Artificial Intelligence*, article 111039. https://doi.org/10.1016/j.engappai.2025.111039

[7] Patrice Lopez. 2009. GROBID: Combining Automatic Bibliographic Data Recognition and Term Extraction for Scholarship Publications. In *Research and Advanced Technology for Digital Libraries: Proceedings of ECDL 2009*. Springer Berlin Heidelberg, Berlin, Heidelberg, 473–474. https://doi.org/10.1007/978-3-642-04346-8_62

[8] Yongqiang Ma, Jiawei Liu, Wei Lu, and Qikai Cheng. 2023. From "what" to "how": Extracting the procedural scientific information toward the metric-optimization in AI. *Information Processing & Management*, 60(3), article 103315. https://doi.org/10.1016/j.ipm.2023.103315

[9] Briony J. Oates. 2006. *Researching Information Systems and Computing*. SAGE Publications, London.

[10] Colin Pilkington and Laurette Pretorius. 2015. A conceptual model of the research methodology domain. In *Proceedings of the 7th International Joint Conference on Knowledge Discovery, Knowledge Engineering and Knowledge Management (IC3K 2015), Volume 2: KEOD*. SCITEPRESS – Science and Technology Publications, Setúbal, Portugal, 96–107. https://doi.org/10.5220/0005613100960107

[11] Wenpeng Yin, Jamaal Hay, and Dan Roth. 2019. Benchmarking Zero-shot Text Classification: Datasets, Evaluation and Entailment Approach. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, Hong Kong, China, November 2019. Association for Computational Linguistics, 3914–3923. https://doi.org/10.18653/v1/D19-1404

Dataset Papers

[D1] Sergey Brin and Lawrence Page. 1998. The anatomy of a large-scale hypertextual web search engine. *Computer Networks and ISDN Systems*, 30(1–7), 107–117. https://doi.org/10.1016/S0169-7552(98)00110-X

[D2] Jeffrey Dean and Sanjay Ghemawat. 2004. MapReduce: Simplified data processing on large clusters. In *Proceedings of the 6th Symposium on Operating Systems Design and Implementation (OSDI '04)*. USENIX Association, 137–150. https://www.usenix.org/conference/osdi-04/mapreduce-simplified-data-processing-large-clusters

[D3] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT 2019)*, Volume 1. Minneapolis, Minnesota: Association for Computational Linguistics, 4171–4186. https://doi.org/10.18653/v1/N19-1423

[D4] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2016)*, 770–778. https://doi.org/10.1109/CVPR.2016.90

[D5] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E. Hinton. 2012. ImageNet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems*, 25, 1097–1105. https://proceedings.neurips.cc/paper_files/paper/2012/hash/c399862d3b9d6b76c8436e924a68c45b-Abstract.html

[D6] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems*, 30, 5998–6008. https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html

Appendix A — Project Roadmap

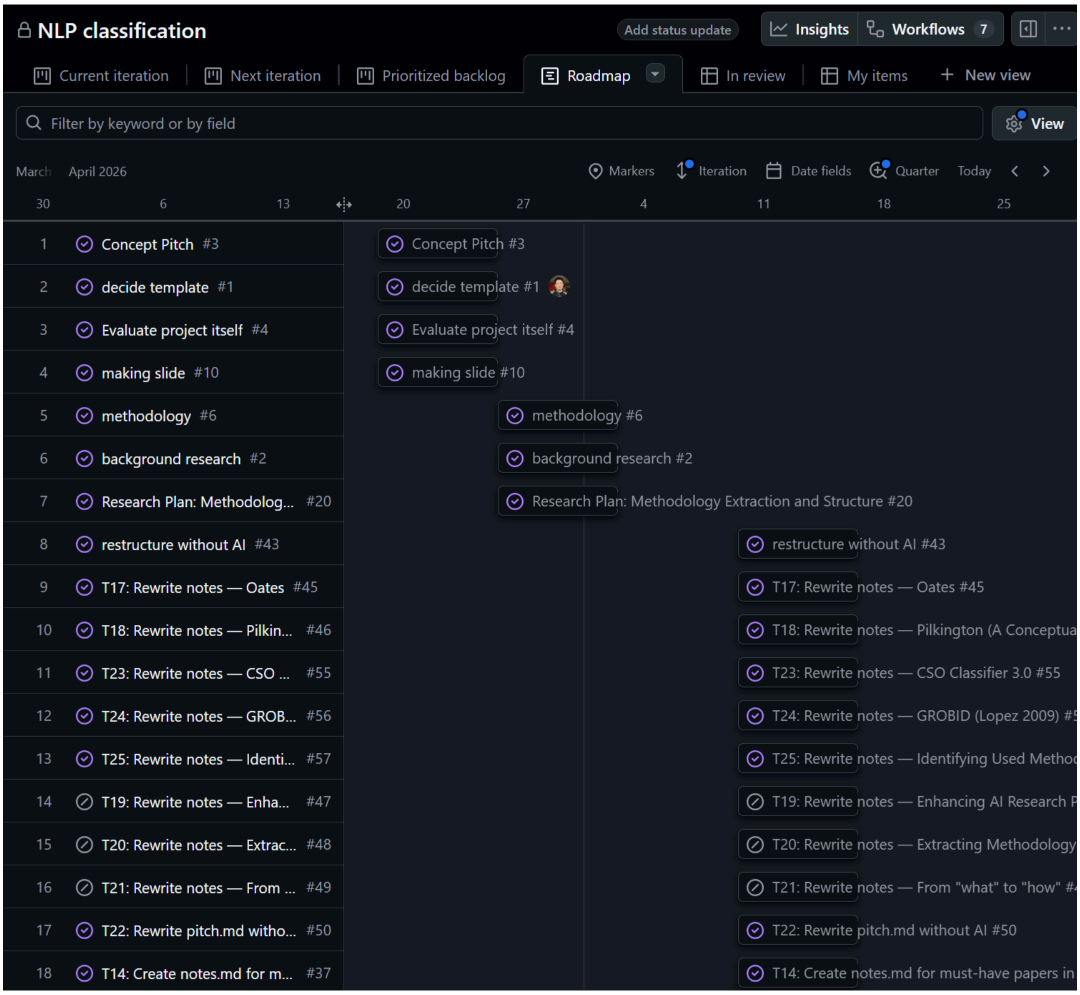

Figure A1: Project roadmap (rows 1–18). Completed tasks from April 2026, including background research, literature notes, and early prototype work.

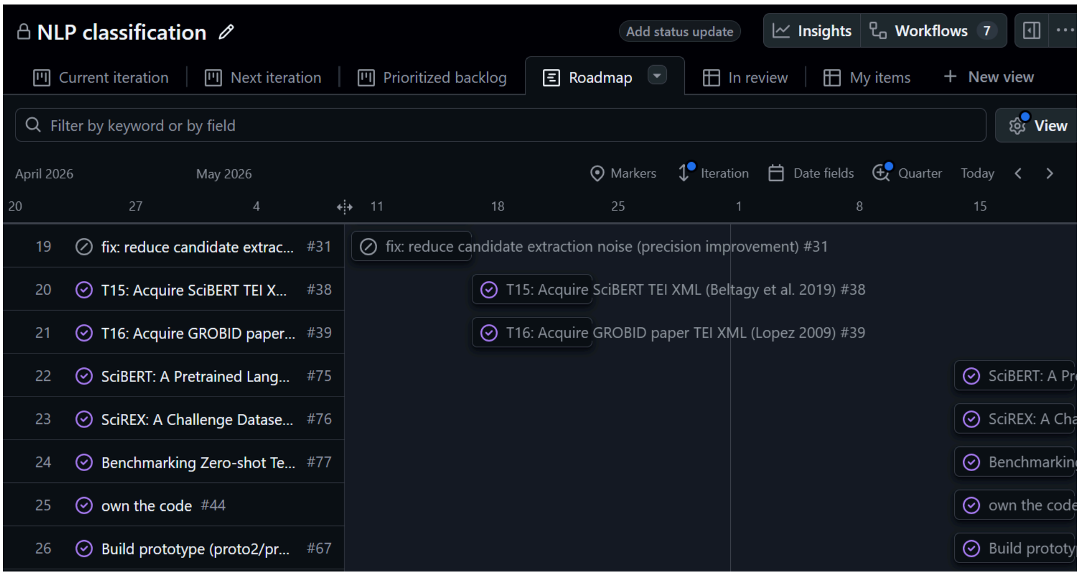

Figure A2: Project roadmap (rows 19–26). April–May 2026, including prototype build, note rewriting, and literature acquisition.

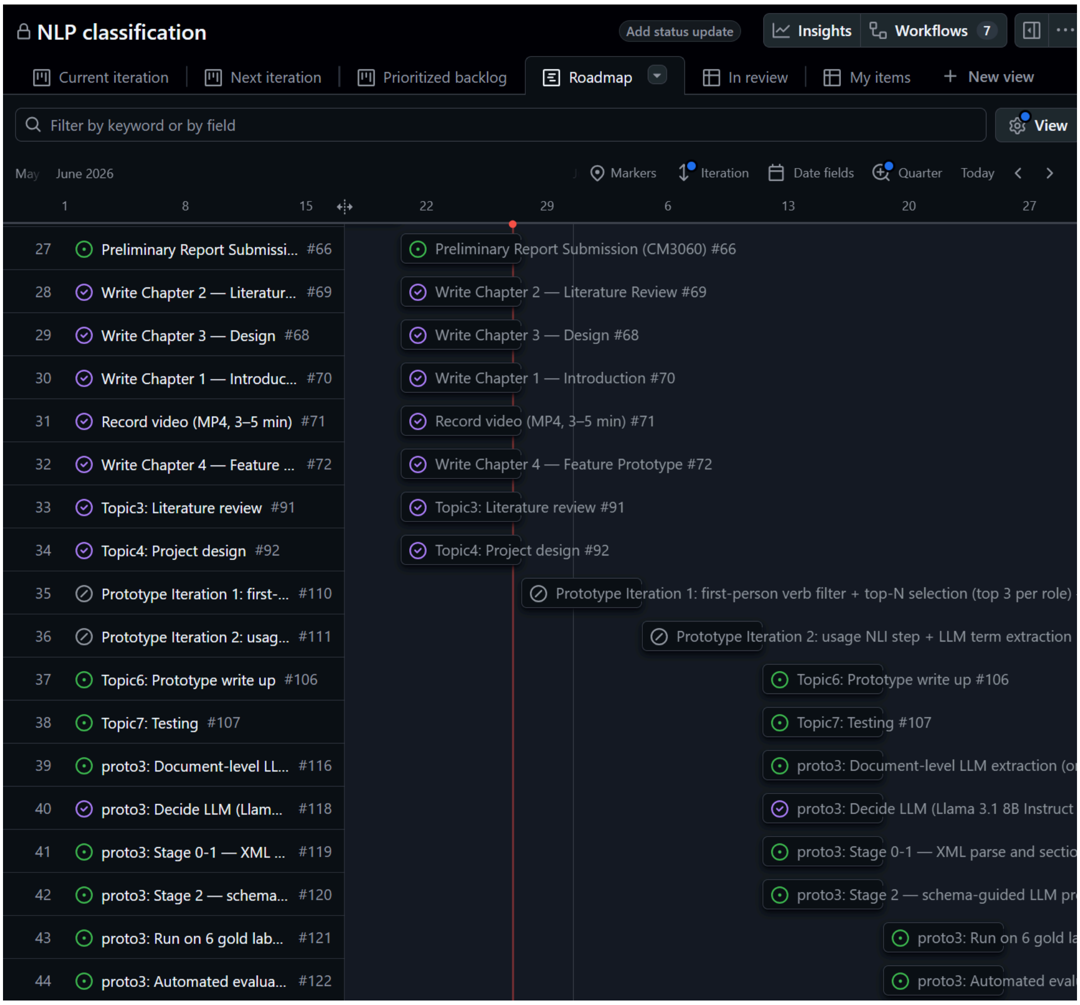

Figure A3: Project roadmap (rows 27–44). June–July 2026, including Preliminary Report submission (row 27, red line), chapter writing, and proto3 tasks (rows 39–44).

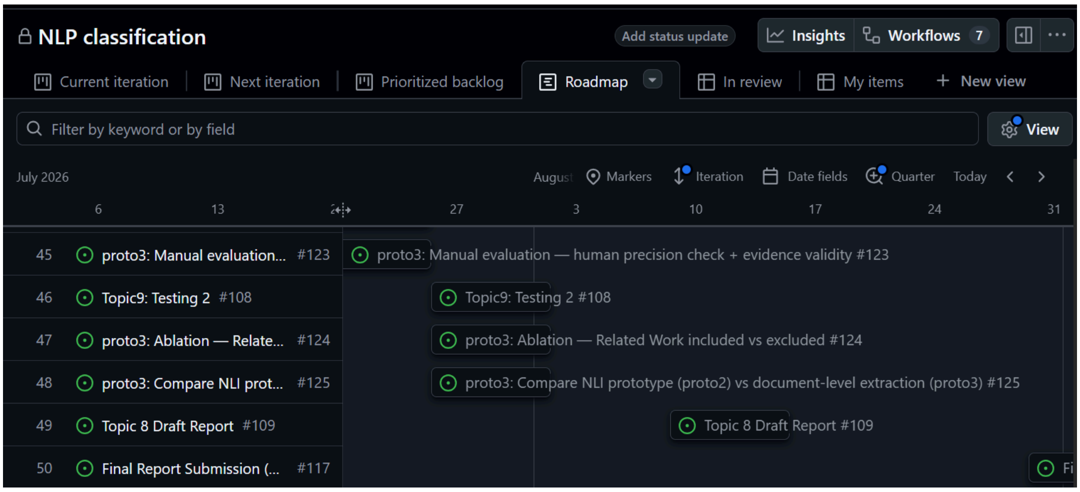

Figure A4: Project roadmap (rows 45–50). July–August 2026, including proto3 evaluation, ablation, comparison, and Final Report submission (Iteration 17).

Appendix B — Extended Evaluation (All 6 Papers)

The primary evaluation in Chapter 4 §3 covers three ML papers (Transformer, BERT, AlexNet) as designed in Chapter 3 §6. This appendix extends the same evaluation to all six dataset papers including two systems papers (MapReduce, Google Search).

Gold labels used:

Paper	TechnicalMethod	Task	Dataset	EvaluationMetric
Transformer	"Transformer"	"machine translation"	"WMT"	"BLEU"
BERT	"BERT"	"GLUE"	"BooksCorpus"	"F1"
AlexNet	"convolutional"	"object recognition"	"ImageNet"	"top-5"
ResNet	"residual"	"image recognition"	"ImageNet"	"top-1"
MapReduce	"MapReduce"	"distributed"	"TeraSort"	"seconds"
Google Search	"PageRank"	"web search"	"million pages"	"quality"

Table B1: Gold labels for all 6 papers.

Results:

Paper	TechnicalMethod	Task	Dataset	EvaluationMetric	Total
Transformer	○	✗	○	✗	2/4
BERT	○	○	○	○	4/4
AlexNet	○	○	○	○	4/4
ResNet	○	✗	○	○	3/4
MapReduce	○	✗	✗	○	2/4
Google Search	✗	○	○	○	3/4
Total					**18/24 (75%)**

Table B2: Extended gold label evaluation results.

ML papers (Transformer, BERT, AlexNet, ResNet) scored 13/16 (81%). Systems papers (MapReduce, Google Search) scored 5/8 (63%). ResNet scored ✗ on Task because "image recognition" does not appear in the 6 accepted Task sentences, likely because the paper frames the task as a competition result rather than an explicit label. MapReduce scored ✗ on Task and Dataset because "distributed" and "TeraSort" are absent from accepted sentences, consistent with the lack of standard ML benchmark structure. Google Search scored ✗ on TechnicalMethod because "PageRank" does not appear in any of the 69 accepted TechnicalMethod sentences, suggesting the algorithm name is mentioned in sentences classified as other roles.